Partial Algebraic Shifting

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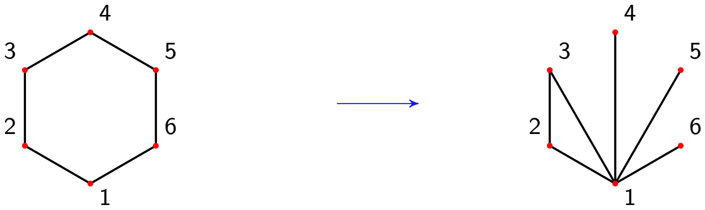
Algebraic Shifting

Definition
$C \in \binom{[n]}{k}$ is <u>shifted</u> if for any $\sigma \in C$, $\sigma|_i^j \in C$ for $i \in \sigma$ and $j < i$.

Theorem (Kalai, Björner-Kalai)
Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f-vector and betti numbers as K.



Exterior Algebraic Shifting

Definition

Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \mathrm{GL}(\mathbb{E}, n)$.

- The kth <u>compound matrix</u> $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$-matrix of k-minors of g.
- $g^{\wedge K_k}$ is the $|K_k| \times \binom{n}{k}$-submatrix of $g^{\wedge k}$ with rows indexed by K_k.

Definition

The <u>partial (exterior) shift</u> of K by g is

$$\Delta_g^{\mathbb{F}}(K) := \bigcup_{k \leq \dim(K)} \left\{ \sigma \in \binom{[n]}{k} : g_{*\sigma}^{\wedge K_k} \notin \mathrm{span}_{\mathbb{E}}\left((g_{*\rho}^{\wedge K_k})_{\rho <_{\mathrm{lex}} \sigma} \right) \right\}.$$

For $g = (x_{ij})_{1 \leq i,j \leq 3}$ and $K = \{\{1,2\},\{2,3\}\}$.

$$g^{\wedge 2} = \begin{pmatrix} x_{11}x_{22} - x_{12}x_{21} & x_{11}x_{23} - x_{13}x_{21} & x_{12}x_{23} - x_{22}x_{13} \\ x_{21}x_{32} - x_{22}x_{31} & x_{21}x_{33} - x_{23}x_{31} & x_{22}x_{33} - x_{23}x_{32} \end{pmatrix},$$

$\Delta(K) = \{\{1,2\},\{1,3\}\}$ mention monte carlo

Algebraic Exterior Shifting

$e_\sigma = e_{\sigma_1} \wedge \ldots \wedge e_{\sigma_k}$ for $\sigma = \{\sigma_1 < \ldots < \sigma_k\}$.

$$I := (e_\sigma : \sigma \notin K) \subset \bigwedge \mathbb{E}^n, I_{\Delta_{g(K)}} = \mathrm{in}(g \cdot I)$$

plucker?

Partial Algebraic Shifting

Consider now $\{\mathfrak{r}(w) \in \mathrm{GL}(\mathbb{E}, n) : w \in S_n\}$.
Let $c_n := (1 \ldots n) \in S_n$.

Theorem (DV-Joswig-Lenzen)

For any simplical complex K and $w \in S_n$ such that $c_n \leq w$ in the weak Bruhat order we have the following,

1. $\Delta_{\mathfrak{r}(w)}(K)$ *is a homotopy equivalent to a wedge of spheres.*
2. $\Delta_{\mathfrak{r}(w)}(K)$ *has the same Betti numbers as* K.

Real Projective Plane

$K = \mathbb{RP}^2_6$, $\mathbb{F} = \mathbb{Q}$

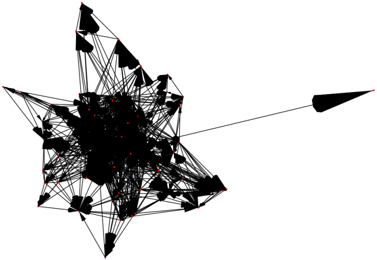


Figure: The partial shift graph starting from $\mathbb{RP}^2_6$, generated using OSCAR.

Matrix decompositions

Lemma

$\Delta_{gb}(K) = \Delta_g(K)$ *for any upper triangular b and any simplicial complex K.*

LU decomposition For $g \in \mathrm{GL}(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu$$

Bruhat Decomposition Let $B \subset \mathrm{GL}(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$\mathrm{GL}(n, \mathbb{E}) = \coprod_{w \in \mathrm{Sym}(n)} BwB \ .$$

Finding the Bruhat cell of g

$$w_0 = (1 \;\; n)(2 \;\; n-1)\dots(\lfloor\frac{n}{2}\rfloor\lceil\frac{n}{2}+1\rceil)$$

If g is generic, then so is $w_0 g$, hence

$$\Delta_{w_0 g}(K) = \Delta_g(K).$$

For the same reason $w_0 g$ there exist some lower triangular l, and upper triangular u', u such that,

$$w_0 g = lu = w_0 u' w_0 u \quad (1)$$

$$g = u' w_0 u \quad (2)$$

$g \in Bw_0 B$

Inversions and Rothe matrix

Definition

For $w \in S_n$, let

$$\begin{aligned} \operatorname{Inv} w &:= \{(i,j) : i < j, i \cdot w > j \cdot w\} \\ \ell(w) &:= |\operatorname{Inv} w| \end{aligned}$$

$$w_0 = \operatorname{argmax}_{w \in S_n} \ell(w)$$

$$\mathfrak{u}(w) := \begin{pmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & \ddots & \ddots & \vdots \\ & & 1 & u_{n-1,n} \\ & & & 1 \end{pmatrix} \quad \text{with} \quad u_{ij} = \begin{cases} x_{ij} & \text{if } (i,j) \in \operatorname{Inv} w, \\ 0 & \text{otherwise,} \end{cases}$$

and let $\mathfrak{r}(w) := \mathfrak{u}(w) w$.

Example

Take $w = (2 \;\; 3)$, $\operatorname{Inv} w = \{(1,2),(1,3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix} = \begin{pmatrix} 1 & y & x \\ 0 & z & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix} \begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix} .$$

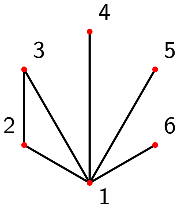
Near cones

Definition

A simplicial complex K is a near cone if for every $\sigma \in K$ with $1 \notin \sigma$ we have $\sigma|_i^1 \in K$ for each $i \in \sigma$.

Lemma (Björner-Kalai 88)

Let K be a near cone. Then K is homotopy equivalent to a wedge of spheres and $\beta_k(K) = |\{\sigma \in K^k : \sigma \cup \{1\} \notin K\}|$.



Shifting to near cones

Lemma (DV-Joswig-Lenzen)

Let K be a simplicial complex on n vertices. Further, let $g \in \mathrm{GL}(n, \mathbb{E})$ be a matrix whose entries in the first column are algebraically independent over $\mathbb{F}$ and the other entries of g. Then $\Delta_g(K)$ is a near cone.

Example

The n-cycle $c_n \coloneqq (1\ 2\ \dots\ n)$ gives rise to the matrix

$$\mathfrak{r}(c_n) = \begin{pmatrix} x_{1,n} & 1 & & \\ \vdots & & \ddots & \\ x_{n-1,n} & & & 1 \\ 1 & 0 & \cdots & 0 \end{pmatrix}.$$

$$\begin{aligned} \Delta^{\mathbb{Q}}_{\mathfrak{r}(c_6)}(\mathbb{RP}^2_6) &= \Delta\{123, 124, 125, 126, 134, 135, 136, 145, 146, 156\} \\ &= \Delta^{\mathbb{Q}}(\mathbb{RP}^2_6) \end{aligned}$$

Shifting to near cones

Lemma (Björner-Kalai)

Let $g \in \mathrm{GL}(n, \mathbb{E})$ be generic and suppose for $\tau \in \binom{[n]}{k}$ there exists $\gamma \in \mathbb{E}$ such that,

$$g_\tau^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma_\sigma g_\sigma^{\wedge K_k}.$$

If $w \in S_n$, then there exists $\gamma' \in \mathbb{E}$ such that

$$g_{\tau \cdot w}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma'_{\sigma \cdot w} g_{\sigma \cdot w}^{\wedge K_k}$$

spell out connection with generic first column shifting to near cone

Weak Bruhat order

- definition and equivalent formulation
- an example with rothe matrix that shows shifting to near cones

Proof of results

Theorem (DV-Joswig-Lenzen)

For K a simplicial complex, $v, w \in S_n$ with $\ell(vw) = \ell(v) + \ell(w)$,

$$\Delta_{\mathfrak{r}(vw)}(K) = \Delta_{\mathfrak{r}(v)\,\mathfrak{r}(w)^v}(K)$$

Mardi slide?

- talk about how file format was useful

QR codes to both papers

Thank You!